

EXAM #1: FALL SEMESTER –PHY2048
18 September 2007 – Instructor: Enrique del Barco

NAME _____

Read carefully the next notes:

- a. Finish reading a problem/question before beginning to solve it.*
- b. For problems: Check the appropriate answer and include the development of the problem in the space below the question. A correct answer without the development of the solution will not be taken into account. Development is not required for questions.*
- c. Only this pages will be taken into account. No extra pages can be used.*
- d. The time for the exam is 60 minutes.*
- e. Gravity acceleration: $g = 9.81 \text{ m/s}^2$*

Grade: (2 problems $\times$ 20 points = 40 points + 12 questions $\times$ 5 points = 60 points) = 100 points

P.1 (20 points): The horizontal and vertical components of the initial velocity of a 0.6-kg football are 16 m/s and 20 m/s respectively. How long does it take for the football to rise to the highest point of its trajectory? What would it be the range, R_m , of the same ball if launched in the moon ($g_{\text{moon}} = 1.6 \text{ m/s}^2$). Give the result in terms of R_{earth} (range of the ball at the Earth's surface).

P.2 (20 points): A 50.0-kg box is being pulled along a horizontal surface by means of a rope that exerts a force of 250 N at an angle of 28.0 degrees above the horizontal. The coefficient of kinetic friction between the box and the surface is 0.350. a) What is the acceleration of the box? b) Above what angle above the horizontal the same force pulling from the box would not be able to move the box if the coefficient of static friction is 0.5?

Q.1 (5 points): If a car accelerates at 4.0 m/s^2 , how long will it take to reach a speed of 80 km/hr, starting from rest?

- A) 1.2 hours B) 5.6 s C) 7.5 s D) 1 min

Q.2 (5 points): Two objects are dropped from a bridge, an interval of 1.0 s apart. During the time that both objects continue to fall, their separation

- A) increases B) increases at first, but then stays constant C) decreases D) is constant

Q.3 (5 points): Which of the following is a scalar quantity?

- A) velocity B) acceleration C) mass D) displacement

Q.4 (5 points): A boy kicks a football with an initial velocity of 20 m/s at an angle of 25 degrees above the horizontal. The acceleration of the ball while it is in flight is

- A) 1.33 m/s² B) 3.66 m/s² C) 1.00 m/s D) 9.81 m/s²

Q.5 (5 points): A student kicks a soccer ball in a high arc toward the opponent's goal. At the highest point in its trajectory:

- A) $\vec{v} = 0$, $\vec{a} = 0$ B) $\vec{v} \neq 0$, $\vec{a} = 0$ C) $\vec{v} = 0$, $\vec{a} \neq 0$ D) $\vec{v} \neq 0$, $\vec{a} \neq 0$

Q.6 (5 points): The horizontal and vertical components of the initial velocity of a football are 24 m/s and 7 m/s respectively. What is the initial velocity of the football?

- A) 31 m/s B) 25 m/s C) 17 m/s D) 42 m/s

Q.7 (5 points): **Q.4 (5 points):** If the initial speed of a projectile is doubled, its range will

- A) double B) quadruple C) be halved D) stay the same

Q.8 (5 points): A force of 120 N is applied to an object whose mass is 30 kg. The object's acceleration is

- A) 4.0 m/s² B) 1.25 m/s² C) 12.0 m/s² D) 75.0 m/s²

Q.9 (5 points): A truck is towing a car whose mass is one quarter that of the truck. The force exerted by the truck on the car is 6000 N. The force exerted by the car on the truck is

- A) 1500 N B) 24000 N C) 6000 N D) 3000 N

Q.10 (5 points): A 40.0-kg crate is being lowered by means of a rope. Its downward acceleration is 2.00 m/s². What is the force exerted by the rope on the crate?

- A) 40 N B) 232 N C) 60 N D) 312 N

Q.11 (5 points): A 39.0-kg child steps on a scale in an elevator. The scale reads 420 N. What is the magnitude of the acceleration of the elevator?

- A) 0.963 m/s² B) 9.81 m/s² C) 46.9 m/s² D) 0.206 m/s²

Q.12 (5 points): When a mass of 25.0 grams is suspended from a certain spring and lowered slowly until the spring stops stretching, the spring stretches 2.00 cm. What is the spring constant of the spring?

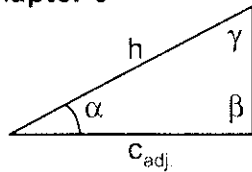
- A) 1.25 N/m B) 0.800 N/m C) 7.85 N/m D) 12.3 N/m

Chapter 2

$$\Delta x = x_f - x_i \quad v_{av} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i} \quad (a = \text{const.}) \quad v_{av} = \frac{v_i + v_f}{2} \quad a_{av} = \frac{\Delta v}{\Delta t}$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad v_x = v_{0x} + a_x t \quad Az^2 + Bz + C = 0 \rightarrow z = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Chapter 3



$$h^2 = c_{adj.}^2 + c_{op}^2 \quad \alpha + \beta + \gamma = 180^\circ$$

$$\sin \alpha = \frac{c_{op.}}{h} \quad \cos \alpha = \frac{c_{adj.}}{h} \quad \tan \alpha = \frac{c_{op.}}{c_{adj.}} = \frac{\sin \alpha}{\cos \alpha}$$

Chapter 4

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad v_x = v_{0x} + a_x t$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \quad v_y = v_{0y} + a_y t$$

Projectile motion: $\text{Range: } x_R = \frac{v_0^2 \sin(2\theta)}{g}$ $\text{Time to Range: } t_R = \frac{2v_0 \sin \theta}{g}$

$\text{Maximum height: } y_M = \frac{v_0^2 \sin^2 \theta}{2g}$ $\text{Time to Max. Height: } t_M = \frac{v_0 \sin \theta}{g}$

$$\vec{v}_{og} = \vec{v}_{og} + \vec{v}_{oc} \quad \vec{v}_{13} = \vec{v}_{12} + \vec{v}_{23} \quad \vec{v}_y = -\vec{v}_{jt}$$

Chapters 5

$$\sum \vec{F} = m\vec{a} \rightarrow \sum F_x = ma_x \quad \sum F_y = ma_y \quad F_{cp} = m \frac{v^2}{r}$$

$$W = mg \rightarrow W_a - W = (\pm)ma \quad \vec{F}_k = \mu_k \vec{N} \quad \vec{F}_{s, \max} = \mu_s \vec{N} \quad \vec{N} = -\sum \vec{F}_i$$